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Fundamentals of Data Science

Unit-2

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PRIMARY DATA COLLECTION

Collecting Original Data Directly from the Source

MEANING

Primary Data Collection is the process of collecting data directly from the original source by the researcher for a specific purpose.

ADVANTAGES

- Highly Accurate
- Original Data
- Specific to the Research Problem
- Up-to-date Information
- Better Decision Making

DISADVANTAGES

- Time Consuming
- Expensive

ARCHITECTURE MODEL



RESEARCHER / DATA SCIENTIST

Identifies the problem and decides what data is needed

↓

CHOOSE DATA COLLECTION METHOD

SURVEY	INTERVIEW	QUESTIONNAIRE	OBSERVATION	EXPERIMENT
Collect data using forms / Google Forms / Online Surveys	Direct interaction with respondents (face-to-face / phone / online)	Set of questions given to respondents to fill and return	Observe behavior or events and record the information	Conduct tests or experiments to collect data under controlled conditions

↓



RESPONDENTS / SOURCES

Students, Customers, Patients, Employees, Users, Farmers, etc.

↓



COLLECTED PRIMARY DATA

Original & First-hand Information

↓

DATA PROCESSING



Data Cleaning →  → Report Generation 

REAL-TIME EXAMPLE

A college conducts a survey asking students,

"How many hours do you study daily?"

The collected responses are primary data.

REAL LIFE APPLICATIONS



Education
Student feedback, study habits survey



Healthcare
Patient health survey, treatment feedback



Business
Customer satisfaction survey, product rating



Banking
Customer opinion, service quality survey



Politics
Election opinion poll, public issues survey

COMPLETE FLOW

Identify Problem → Decide Data Needed → Choose Method → Collect Data from Respondents → Verify & Store Data → Analyze Data → Report & Decision Making

UNIT–II : DATA COLLECTION AND PREPROCESSING (PART – 1)

Secondary Data Collection | Data Sources | Comparison of Data Sources

1. SECONDARY DATA COLLECTION

Meaning

Data already collected by someone else and reused for another purpose.

Examples

- Government Census
- Kaggle Datasets
- WHO Reports
- RBI Reports
- Company Annual Reports

Advantages

- Less Expensive
- Saves Time
- Easily Available
- Large Amount of Data

Disadvantages

- May be Outdated
- May not match the research objective
- Unknown Accuracy

Flow of Secondary Data Collection

Existing Sources (Collected by others)
↓
Download Dataset
↓
Data Cleaning
↓
Machine Learning / Analysis
↓
Insights / Decisions

Real-Time Example

A student downloads the Titanic Dataset from Kaggle for machine learning practice.



2. DATA SOURCES

A. STRUCTURED DATA

Meaning

Data stored in rows and columns.

Examples

- Student Database
- Employee Records
- Banking Database

Example (Table)

Roll No	Name	Marks
101	Rahul	85
102	Anjali	90
103	Ravi	78

Stored in relational databases (SQL)

B. SEMI-STRUCTURED DATA

Meaning

Data does not follow a strict table format but has tags or labels.

Examples

- XML
- JSON
- HTML

Example (JSON)

```
{
  "Name": "Rahul",
  "Age": 20
}
```

Flexible format (Tag based)

C. UNSTRUCTURED DATA

Meaning

Data without any predefined structure.

Examples

- Images
- Videos
- Audio
- Emails
- Social Media Posts

No fixed format (Free form data)

3. COMPARISON OF DATA SOURCES

Aspect	Structured Data	Semi-Structured Data	Unstructured Data
Structure	Rows & Columns	Tags	No Format
Format Example	Table, Excel, SQL	XML, JSON, HTML	Images, Videos, Audio, Text
Storage	SQL Database	NoSQL, JSON Files	File System, Data Lake
Ease of Analysis	Easy	Moderate	Difficult
Schema	Fixed Schema	Flexible Schema	No Schema
Tools	SQL, Excel	JSON, XML Parsers	AI, NLP, CV Tools

Memory Trick

Structured = Tables | Semi-Structured = Tags | Unstructured = No Format

Key Takeaway

Secondary data helps us save time, money and effort. Data can be in different formats like tables, tags or media files. Choose the right source based on your research need.

UNIT–II : DATA COLLECTION AND PREPROCESSING (PART – 2)

Data Collection Techniques | Challenges in Data Collection | Solutions

4. DATA COLLECTION TECHNIQUES

Surveys



Collect opinions using questionnaires.

Example:
Google Forms Feedback

Interviews



Direct interaction with respondents.

Example:
HR interviews job candidates.

Observation



Watching behavior without asking questions.

Example:
A supermarket records customer movement using CCTV.

Sensors



Collect data automatically.

Example:
Temperature sensors in weather stations.

Web Scraping



Collecting data from websites.

Example:
Price comparison websites collect prices from Amazon & Flipkart.

APIs



APIs provide data directly.

Example:
Weather API provides live temperature data.

Data Collection Flow (General)



5. CHALLENGES IN DATA COLLECTION

Missing Data	:	Some values are not available.
Duplicate Data	:	Same data recorded multiple times.
Incorrect Data	:	Wrong or inconsistent information.
Privacy Issues	:	Personal data may be misused.
Data Security	:	Data may be lost, hacked or accessed without permission.



Why Challenges Occur?

- Human error
- Large volume of data
- Different data formats
- Security vulnerabilities

6. SOLUTIONS

Data Validation



Check data correctness at every step.

Remove Duplicates



Eliminate repeated records.

Accuracy Check



Verify and correct errors using rules and constraints.

Data Privacy & Consent



Take permission and protect personal data.

Encryption & Backup



Secure data and keep regular backups.

7. KEY TAKEAWAY



- Data can be collected using various techniques.
- Data comes in different formats: Structured, Semi-Structured and Unstructured.
- Challenges must be handled carefully to ensure high-quality data.
- Good quality data is the foundation of accurate analysis and better decisions.



Better Data
Better Analysis
Better Decisions

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UNIT-II : DATA CLEANING (PART - 1)

"Clean Data → Accurate Analysis → Better Decisions"

1. MEANING

Data cleaning means identifying and correcting errors, inconsistencies, and unwanted information from a dataset. It improves the quality of the data before analysis.

2. DEFINITION

Data Cleaning is the process of detecting, correcting, or removing inaccurate, incomplete, duplicate, or inconsistent data.

3. WHY DATA CLEANING?

Poor quality data leads to

- Wrong predictions
- Poor accuracy
- Incorrect business decisions

Hence,

"Garbage In = Garbage Out (GIGO)
If bad data is given as input, the output will also be poor.

4. COMMON DATA PROBLEMS

1. MISSING VALUES

Example

Name	Age
Rahul	20
Anjali	NA

Information not available.

2. DUPLICATE RECORDS

Example

ID	Name
101	Rahul
101	Rahul

Same record appears more than once.

3. INCORRECT DATA

Example

Age = -5

Impossible value.

Contains impossible or invalid values.

4. INCONSISTENT DATA

Example

Male
male
MALE

Same category in different formats. Should be standardized.

5. WHY DO WE NEED DATA CLEANING?

Real-Time Example:

A hospital database contains a patient's temperature = 450°C. The AI system may predict an impossible disease because the input itself is incorrect.

6. DATA CLEANING ARCHITECTURE

RAW DATA	DATA CLEANING	CLEAN DATASET	ANALYSIS / ML MODEL
• Missing • Duplicate • Incorrect • Inconsistent • Outliers	• Remove Duplicates • Handle Missing Values • Correct Errors • Standardize Values • Remove Invalid Data • Detect Outliers	• Accurate • Consistent • Complete • Reliable	• Better Accuracy • Better Predictions • Better Decisions

17. STEPS IN DATA CLEANING

1. Remove Duplicates
2. Handle Missing Values
3. Correct Inconsistent Values
4. Remove Invalid Data
5. Detect Outliers
6. Verify Final Dataset

8. BENEFITS OF DATA CLEANING

- Improves data quality
- Increases model accuracy
- Better predictions
- Supports correct business decisions
- Saves time and cost

9. REAL-TIME EXAMPLES

1. COLLEGE

Same student entered multiple times in attendance register.

→ Remove duplicate records.

2. BANKING

Customer age entered as 250. Impossible value.

→ Correct or remove record.

3. E-COMMERCE

Product names in different formats. (iPhone, iphone, IPHONE)

→ Standardize to one format.

4. HOSPITAL

Temperature entered as 450°C. Incorrect data.

→ Correct to real value.

5. SURVEYS

Respondent skipped salary question in Google Form.

→ Handle missing value.

10. MEMORY TIP

Clean the data before you analyze the data.
Good data → Good model → Good results.

Remove | Handle | Correct | Validate
= Clean Data

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UNIT-II : DATA CLEANING (PART - 2)

Process → Quality → Reliable Results

1. DATA CLEANING PROCESS FLOW

Raw Dataset
Collect data from various sources.

↓

Remove Duplicate Records
Identify and remove duplicate entries.

↓

Handle Missing Values
Fill or replace missing data using appropriate methods.

↓

Correct Inconsistent Values
Standardize text, categories, formats (e.g., Male, male, MALE → Male).

↓

Remove Invalid Data
Delete or correct impossible or meaningless values.

↓

Detect Outliers
Identify values that are too high or too low compared to others.

↓

Verify Final Dataset
Check for any remaining issues and validate the dataset.

↓

Clean Dataset Ready
Data is now ready for analysis or machine learning model.

2. DETAILS OF EACH STEP

Step	Purpose	Purpose	Key Actions	Example
1	Remove Duplicates	Avoid repetition of the same data.	Check based on ID or key columns and remove.	Before: ID, Name — 101 Rahul; 101 Rahul. After: ID, Name — 101 Rahul
2	Handle Missing Values	Fill missing data appropriately.	Use mean, median, mode, forward fill, etc.	Before: Name, Salary — Rahul 50000; Priya NA. After: Name, Salary — Rahul 50000; Priya 48000
3	Correct Inconsistent Values	Make data uniform and consistent.	Standardize text, case, formats, units.	Before: Gender — Male; male; MALE. After: Gender — Male; Male; Male
4	Remove Invalid Data	Remove incorrect or meaningless data.	Delete negative age, impossible values, empty strings, etc.	Before: Age — 25; -5; 30. After: Age — 25; 30
5	Detect Outliers	Improve model accuracy.	Use methods like IQR, Z-score, Boxplot.	Outliers → 0 10 20 30 40 50 60 70 80 90 100
6	Verify Final Dataset	Ensure data is clean, correct and ready.	Check for duplicates, missing values, invalid data, outliers.	All checks passed. Dataset is ready!

3. REAL-TIME CASE STUDY – COLLEGE PLACEMENT ANALYSIS

RAW DATA (PROBLEMS)

Student	CGPA	Package (LPA)
Rahul	8.5	4.0
Rahul	8.5	4.0
Priya	NA	3.5
Kiran	15.2	6.0
Anjali	-1.0	NA

- Duplicate record (Rahul)
- Missing CGPA (Priya)
- Invalid CGPA (Kiran = 15.2, Anjali = -1.0)
- Missing Package (Anjali)

DATA CLEANING ACTION

- Removed duplicate record of Rahul
- Filled missing CGPA of Priya using mean (Example: 8.1)
- Corrected invalid CGPA: Kiran (15.2 → 9.5), Anjali (-1.0 → 7.0)
- Filled missing Package of Anjali using mean (Example: 4.4)

CLEAN DATA (AFTER CLEANING)

Student	CGPA	Package (LPA)
Rahul	8.5	4.0
Priya	8.1	3.5
Kiran	9.5	6.0
Anjali	7.0	4.4

Clean data leads to better analysis and accurate placement prediction.

4. DENEFITS OF DATA CLEANING

- Improves data quality
- Increases model accuracy
- Better predictions
- Supports correct business decisions
- Saves time and cost
- Builds trust in data

5. REAL-TIME EXAMPLES

- **College:** Remove duplicate attendance records before calculating percentage.
- **Banking:** Correct invalid ages and fill missing income details.
- **E-commerce:** Standardize product names (iPhone, iphone, IPHONE → iPhone).
- **Hospital:** Correct wrong temperature entries and handle missing test results.
- **Surveys:** Remove incomplete or duplicate survey responses.

6. MEMORY TIP

Clean the data before you analyze the data.
Good Data → Good Model → Good Results

Raw Data → Clean Data → Accurate Analysis → Better Decisions → Success

"Clean Today, Accurate Tomorrow!"

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UNIT-II : HANDLING MISSING VALUES (PART - 3)

Handle Missing Data → Improve Quality → Better Predictions

1. MEANING

Missing values occur when some information is not recorded or unavailable.

Example:

Name	Salary
Rahul	50000
Priya	NA

Here, Priya's salary is missing.

2. WHY MISSING VALUES OCCUR?

- User skipped the question.
- Sensor failure.
- Database error.
- Data corruption.
- Manual entry mistakes.

3. TYPES OF MISSING VALUES

MCAR – Missing Completely At Random

Missing values occur randomly and are unrelated to other variables.

Example: Internet failure caused random missing records.

MAR – Missing At Random

Missing values depend on another observed variable.

Example: Women skipped salary information more often than men.

MNAR – Missing Not At Random

Missing values are related to the missing value itself.

Example: High-income people intentionally do not disclose salary.

4. METHODS TO HANDLE MISSING VALUES

1. DELETE ROWS

Remove entire rows that have missing values.

Use when very few records are missing.

2. DELETE COLUMNS

Remove columns that contain too many missing values.

Use when a column has more than 50–70% missing data.

3. MEAN IMPUTATION (Numerical Data)

Replace missing value with the average (mean).

Example:

Marks
70
80
NA
90

Mean (70+80+90)/3 = 80

Missing value becomes 80.

4. MEDIAN IMPUTATION (Numerical Data)

Replace missing value with the median.

Useful when data contains outliers.

5. MODE IMPUTATION (Categorical Data)

Replace missing value with the most frequent value (mode).

Example:

Color
Red
Blue
Red
NA

Mode = Red

Replace NA with Red.

6. FORWARD FILL (Last Observation Carried Forward)

Use the previous non-missing value to fill the missing value.

Best for time-series data.

Example:

Day	Temperature (°C)	After Forward Fill
Day 1	30	30
Day 2	31	31
Day 3	NA	31
Day 4	32	32

7. BACKWARD FILL (Next Observation Carried Backward)

Use the next non-missing value to fill the missing value.

Best for time-series data.

Example:

Day	Temperature (°C)	After Backward Fill
Day 1	30	30
Day 2	NA	32
Day 3	NA	32
Day 4	32	32

5. ADVANTAGES

- Preserves dataset size.
- Reduces information loss.
- Improves model performance.
- Helps in better predictions.
- Prepares data for analysis and ML models.

6. LIMITATIONS

- May introduce bias if wrong method is chosen.
- Can hide the real pattern in data.
- Too much missing data reduces quality.
- Incorrect handling can reduce model accuracy.

7. MISSING VALUE HANDLING FLOW

Dataset with Missing Values → Identify Missing Values → Choose Suitable Method → Impute / Handle Missing Values → Clean Dataset Ready for Analysis

8. SUMMARY TABLE

Method	Type of Data	How it Works	When to Use	Example
Delete Rows	Numerical / Categorical	Remove rows with missing values	Few missing rows	Remove 2 records out of 1000
Delete Columns	Numerical / Categorical	Remove column with many missing values	>50% missing in a column	Drop 'Phone Number' column
Mean Imputation	Numerical	Replace with average	Data is normally distributed	Mean of marks = 80
Median Imputation	Numerical	Replace with median value	Data has outliers	Median salary = 45000
Mode Imputation	Categorical	Replace with most frequent value	Categorical data	Replace NA with 'Red'
Forward Fill	Time-Series	Use previous value	When trend changes slowly	Temperature Day 3 = Day 2
Backward Fill	Time-Series	Use next value	When next value is more reliable	Temperature Day 2 = Day 3

9. MEMORY TIP

Delete → Mean → Median → Mode → Forward → Backward

Remember: Choose the best method based on data type and situation.

Good Data → Good Model → Good Results !

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4. DATA TRANSFORMATION – PART 1

INTRODUCTION & BASICS

1. MEANING

Data Transformation means converting data from one format, structure, or scale into another format that is more suitable for analysis and machine learning.

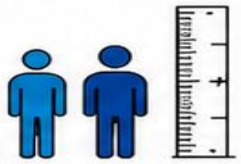
2. DEFINITION

Data Transformation is the process of converting raw data into a suitable format so that it becomes easier to analyze, compare, visualize, and use in machine learning algorithms.

3. SIMPLE ANALOGY

HEIGHT EXAMPLE

Rahul = 170 cm
John = 1.70 m
Ravi = 1700 mm



All represent the same height but in different units.
Converting them into the same unit is **Data Transformation.**

CURRENCY EXAMPLE

India → Rupees (₹)
USA → Dollars ($)
Japan → Yen (¥)

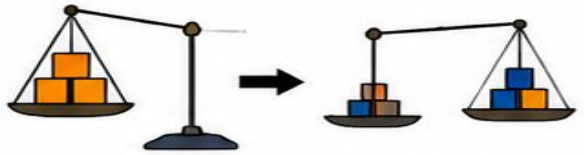


To compare prices, we convert all currencies into one currency.
This conversion is also **Data Transformation.**

4. WHY DO WE NEED DATA TRANSFORMATION?

WITHOUT TRANSFORMATION

- Features have different scales
- Difficult to compare values
- Poor machine learning performance
- Slow convergence
- Lower prediction accuracy



Unbalanced (Raw Data) → Balanced (Transformed Data)

WITH TRANSFORMATION

- Equal contribution of features
- Faster training
- Better accuracy
- Better visualization
- Better predictions

5. OBJECTIVES

- Convert data into a common format
- Improve data consistency
- Make numerical values comparable
- Improve data visualization
- Increase machine learning accuracy
- Reduce computational complexity
- Improve algorithm convergence

KEY POINT: Data Transformation does not remove data. It only changes the format, structure, or scale of clean data so that it becomes suitable for analysis and machine learning.

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4. DATA TRANSFORMATION – PART 2

WHY TRANSFORM? & REAL-TIME EXAMPLES

1. RAW DATA EXAMPLE

Feature	Value
Age	25
Salary	₹500000
Height	170

Salary values are **20,000 times** larger than Age values.

ML algorithms may give more importance to Salary.

2. AFTER TRANSFORMATION

Feature	Transformed Value
Age	0.52
Salary	0.61
Height	0.49

Now all features contribute fairly to the machine learning model.

3. REAL-TIME EXAMPLES

BANKING

Customer Details

Customer	Age	Salary (₹)
Rahul	25	35,000
Priya	45	8,50,000
Kiran	35	15,00,000

Salary values dominate the dataset. Transformation makes features comparable for credit scoring.

HEALTHCARE

Patient Details

Patient	Weight (kg)	Height (cm)	Sugar (mg/dL)
Rahul	75	170	250
Priya	60	160	190
Kiran	80	180	210

Different units (kg, cm, mg/dL). Transformation converts them into comparable values for better diagnosis.

E-COMMERCE

Customer Data

Customer	Age	Income (₹)	Score
A	24	12,00,000	85
B	30	6,00,000	65
C	22	3,00,000	40

Income dominates because of large values. Transformation balances all features for better recommendations.

WEATHER FORECASTING

Weather Data

Temp (°C)	Humidity (%)	Rain (mm)
32	85	12.5
28	70	8.0
35	90	20.0

Different scales affect prediction. Transformation improves the performance of weather prediction models.

EDUCATION

Student Marks From Different Exams

Exam	Max Marks	Student Marks
School Exam	100	85
Board Exam	500	350
Entrance Exam	1000	720

Different exam scales are converted to a common scale for fair comparison and ranking.

KEY POINT: Data Transformation helps in converting data into a common format, so that machine learning models treat all features equally and produce accurate results.

- Makes data comparable
- Improves model performance
- Essential for many ML algorithms

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4. DATA TRANSFORMATION – PART 3

ARCHITECTURE, FLOW & BENEFITS

1. ARCHITECTURE DIAGRAM

RAW DATA

Age	= 25
Salary	= ₹500000
Height	= 170
Gender	= Male

↓

DATA TRANSFORMATION

- Remove Unit Differences
- Scale Features
- Convert Categories
- Normalize Values

↓

TRANSFORMED DATA

Age	= 0.45
Salary	= 0.62
Height	= 0.58
Gender	= 1

↓

MACHINE LEARNING MODEL



2. DATA TRANSFORMATION FLOW



3. BENEFITS OF DATA TRANSFORMATION

- Improves data quality and consistency
- Makes data suitable for machine learning algorithms
- Reduces the impact of varying feature scales
- Enhances prediction accuracy
- Speeds up model training
- Simplifies data visualization
- Supports fair comparison between variables
- Improves algorithm convergence
- Helps in better decision making

MEMORY TIP

Clean Data → Transform Data → Better Model → Better Results

Transformation makes data comparable, consistent and suitable for machine learning.

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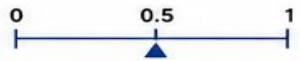
4. DATA TRANSFORMATION – PART 4

TECHNIQUES OVERVIEW & APPLICATIONS

1. COMMON TECHNIQUES

1 Normalization

Converts numerical values into a fixed range (usually 0 to 1).



Formula: $X' = \frac{X - X_{min}}{X_{max} - X_{min}}$

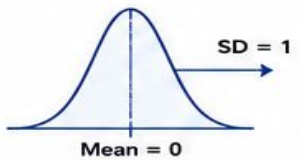
Applications: Neural Networks, KNN, Image Processing, Deep Learning.

Example (Marks)	
Marks	Normalized Value
40	0.00
60	0.33
70	0.50
80	0.67
100	1.00

2 Standardization

Transforms values so that the dataset has Mean = 0 and Standard Deviation = 1.

Formula: $Z = \frac{X - \mu}{\sigma}$



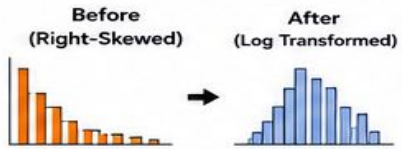
Example (Salary in ₹)	
Salary (₹)	Z-score
40,000	-1.22
50,000	0.00
60,000	1.22
70,000	2.45

Applications: Linear Regression, Logistic Regression, SVM, PCA, Clustering.

3 Log Transformation

Reduces very large values and right-skewed data using logarithm.

Formula: $X_{new} = \log(X)$ (usually natural log)



Example (Income in ₹)	
Income (₹)	Log(Income)
20,000	9.90
30,000	10.31
40,000	10.60
15,00,000	14.22

Applications: Financial data, Population statistics, Sales, Income analysis.

4 Encoding

Converts categorical (text) data into numerical form.

a) Label Encoding	
Category	Encoded Value
Male	1
Female	0

b) One-Hot Encoding			
Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Applications: Logistic Regression, Decision Trees, Naïve Bayes, Neural Networks.

5 Binning

Groups continuous numerical values into intervals (bins).

Example (Marks → Grades)	
Marks (Range)	Grade
0 – 39	Fail
40 – 59	Pass
60 – 79	First Class
80 – 100	Distinction

Example (Age → Category)	
Age (Years)	Category
0 – 17	Young
18 – 35	Adult
36 – 60	Middle Age
60+	Senior

Applications: Customer segmentation, Risk analysis, Education grading, Surveys.

2. REAL-TIME APPLICATIONS

Banking	Normalize customer income before credit scoring.	Example: Income (₹20k – ₹20L) normalized to 0 – 1.
Healthcare	Standardize patient indicators (e.g., BP, Cholesterol, Glucose).	Example: BP (120–180 mmHg) standardized for comparison.
E-Commerce	Encode product categories and customer preferences.	Example: Electronics = 1, Clothing = 2, Home = 3.
Education	Normalize marks from different examinations before rankings.	Example: School (100) and Board (500) scaled to 0 – 1.
Weather Forecasting	Standardize temperature, humidity, rainfall for prediction models.	Example: Temp (10–40°C) standardized for ML model.
Social Media	Encode user interests and transform text features for sentiment analysis.	Example: Interest "Sports" = 1, "Movies" = 2, "Travel" = 3.

3. SUMMARY

Raw Data (Different Formats & Scales)
↓
Clean Data (Remove Errors, Missing Values)
↓
Transform Data (Normalize, Standardize, Encode, Bin, Log Transform)
↓
Machine Learning Model (Improved Performance)
↓
Better Predictions & Insights

Technique	Purpose	Best Used For
Normalization	Scale values between 0 and 1	KNN, Neural Networks, Image Processing
Standardization	Mean = 0, SD = 1	Regression, SVM, PCA, Clustering
Log Transformation	Reduce skewness & large values	Financial, Sales, Income, Population
Encoding	Convert categories to numbers	All ML Algorithms
Binning	Group values into intervals	Segmentation, Grading, Risk Analysis

4. KEY BENEFITS

- Improves accuracy of ML models
- Speeds up model training
- Ensures fair comparison of features
- Improves data visualization
- Makes data consistent & machine-readable

KEY POINT: Data Transformation is essential to convert cleaned data into a consistent, comparable, and machine-readable format. It enhances model performance, efficiency, and helps in making accurate predictions in real-world applications.

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PART 1

5. FEATURE ENGINEERING

1. INTRODUCTION

Feature Engineering is the process of **creating, modifying, selecting,** or **transforming** existing data into **more useful features** that improve data analysis and machine learning **performance.**

2. EXAMPLE

Raw Data

Name	Date of Birth	Marks
Rahul	15-05-2005	420

Engineered Features

Age	Percentage	Passed
20	84%	Yes

New features provide more meaningful information for analysis and help improve prediction accuracy.

3. WHY DO WE NEED FEATURE ENGINEERING?

IMPROVES ACCURACY
Creates better features that help models predict more accurately.

DISCOVERS HIDDEN PATTERNS
Helps in finding hidden relationships and valuable insights.

SPEEDS UP MODEL LEARNING
Well-engineered features make the model learn faster and better.

4. ARCHITECTURE OF FEATURE ENGINEERING

RAW DATA
Unprocessed data from various sources.

→

FEATURE ENGINEERING
- Create New Features
- Select Useful Features
- Transform & Encode
- Extract Information

→

SMART FEATURES
Clean, meaningful and model-ready features.

→

ML MODEL
Model learns better from smart features.

→

BETTER PREDICTION
Improved accuracy and reliable predictions.

5. OBJECTIVES

- Create meaningful features from raw data.
- Select only useful and important features.
- Convert raw data into informative variables.
- Improve accuracy and reduce complexity.
- Help models learn better and faster.

DIFFERENCE BETWEEN DATA TRANSFORMATION & FEATURE ENGINEERING

DATA TRANSFORMATION	FEATURE ENGINEERING
Standard methods like normalization, standardization, encoding, etc.	Creates or modifies features based on domain logic, hidden relationships or business understanding.
Focuses on preparing data for machine learning.	Focuses on making data more informative and useful for the model.
Changes the scale or format of existing data.	Creates new meaningful variables or extracts useful information.
Example: Scaling, Encoding, Handling Missing Values.	Example: Age from DOB, Loan-to-Income Ratio, Total Spending, Festival Season.

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PART 2

5. FEATURE ENGINEERING TECHNIQUES

A. KEY FEATURE ENGINEERING TECHNIQUES

Technique	What It Does	Example (Input → Output)	Use Case / Benefit
1 Feature Creation	Creates new features from existing ones using logic or calculations.	Input : Date of Birth = 15-05-2005 Output: Age = 20 (in 2025)	✓ Helps models understand better and improve accuracy.
2 Feature Selection	Selects only important features and removes irrelevant or redundant ones.	Input : Age, Salary, Phone No., Address, Loan Amount Output: Age, Salary, Loan Amount	✓ Reduces complexity and improves model performance.
3 Feature Extraction	Extracts useful information from raw data.	Input : Address = "Vizag, Andhra Pradesh, India" Output: City = Vizag State = Andhra Pradesh Country = India	✓ Converts unstructured data into structured meaningful features.
4 Encoding (Categorical Data)	Converts categorical/text data into numerical values.	Input : Gender = Male Output: Male = 1, Female = 0	✓ Makes categorical data usable for machine learning models.
5 Date-Time Features	Creates useful time-based features from date/time information.	Input : Date = 15-Aug-2025 Output: Day = Friday Month = August Is_Weekend = No	✓ Captures hidden patterns related to time, season, or events.
6 Text Features	Converts text data into useful numerical features.	Input : Review = "Very good product" Output: Positive_Score = 0.92 Words_Count = 3	✓ Helps in sentiment analysis and NLP applications.

B. REAL-TIME EXAMPLES OF FEATURE ENGINEERING

1. BANKING EXAMPLE – Loan Approval Prediction

Raw Features	Engineered Features
Age, Salary, Loan Amount	Loan-to-Income Ratio
Date of Application	Months Since Application
Marital Status	Marital Status Encoded (0/1)
Employment Type	Is Self Employed? (0/1)

2. EDUCATION EXAMPLE – Student Performance

Raw Features	Engineered Features
Date of Birth	Age
Total Marks	Percentage
Subject Marks	Average Marks per Subject
Attendance %	Is Regular? (0/1)

3. HEALTHCARE EXAMPLE – Disease Prediction

Raw Features	Engineered Features
Weight (kg), Height (cm)	BMI (Body Mass Index)
Blood Pressure (Systolic)	BP Category (High/Normal)
Date of Checkup	Months Since Last Checkup
Glucose Level	Glucose Level Category

C. FEATURE ENGINEERING FLOW



D. OBJECTIVES

- Create meaningful and useful features.
- Select only important features.
- Transform data into informative variables.
- Improve accuracy and reduce complexity.
- Help models learn better and faster.

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E. ADVANTAGES

- Improves prediction accuracy.
- Reduces data complexity.
- Speeds up model training.
- Handles real-world data better.
- Reveals hidden patterns.

F. MEMORY TIP

C → Create features
S → Select useful features
E → Extract useful info
E → Encode data
D → Date-time features
T → Text features

Remember: "CSEEDT" for Feature Engineering!

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PART 3

6. FEATURE SCALING

1. MEANING & DEFINITION

Meaning:

Feature Scaling is the process of adjusting numerical features to a common scale so that all features contribute equally during model training.

Definition:

Feature Scaling is a preprocessing technique that transforms numerical features into the same range or distribution, allowing machine learning algorithms to treat all features equally.

2. WHY IS FEATURE SCALING REQUIRED?

Example Without Scaling

Feature	Value
Age	25
Salary (₹)	500000
Height (cm)	170

Without scaling, the algorithm may focus more on Salary because of its larger values.

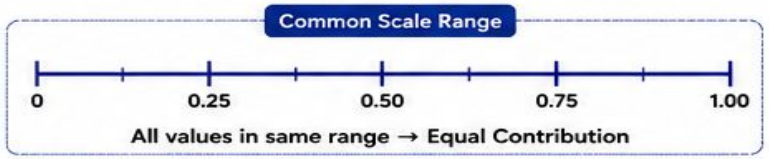
Example With Scaling (0 to 1)

Feature	Scaled Value
Age	0.48
Salary (₹)	0.53
Height (cm)	0.50

After scaling, all features have similar ranges and equal importance.

3. ARCHITECTURE OF FEATURE SCALING

Raw Features (Different Scales) → Feature Scaling (Transform) → Scaled Features (Common Scale) → ML Algorithm (Model Training) → Better Performance & Accurate Results



4. TYPES OF FEATURE SCALING

- Min-Max Scaling (Normalization)
- Standard Scaling (Z-Score Scaling)
- Robust Scaling
- Max Absolute Scaling
- Unit Vector Scaling

5. REAL-TIME APPLICATIONS

Banking	Used in credit scoring, loan approval prediction, and fraud detection.
Healthcare	Helps in disease prediction, patient risk analysis, and treatment planning.
E-Commerce	Improves recommendation systems, customer segmentation, and sales forecasting.
Education	Used in student performance analysis and placement prediction.
Weather Forecasting	Improves prediction models for temperature, rainfall, and storms.

6. ADVANTAGES

- Faster convergence of ML algorithms.
- Improved prediction accuracy.
- Prevents features with large values from dominating.
- Helps distance-based algorithms (KNN, SVM, K-Means).
- Better performance and stability of models.
- Equal importance to all features during training.

7. SUMMARY

- Feature Scaling ensures that all numerical features are on the same scale.
- It improves learning speed, accuracy, and overall performance of machine learning models.
- Always apply scaling after data cleaning and before model training.
- Choose the right scaling method based on data distribution and problem type.

8. MEMORY TIP

S (Same Scale) → C (Contributes Equally) → A (Algorithm Works Better) → L (Learning Faster) → E (Enhanced Accuracy)

Think: SCALE = Smart Calculation of All Leverages Equally

KEY POINT: Feature Scaling balances the importance of all numerical features by putting them on a common scale, which leads to better model performance and faster training.

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PART 4

6. TYPES OF FEATURE SCALING

METHODS, FORMULAE, COMPARISON & APPLICATIONS

1 MIN-MAX SCALING (NORMALIZATION)

Transforms features to a fixed range [0, 1].

- Range / Output: 0 to 1
- Best Used For: KNN, Neural Networks, Image Processing
- Sensitive to Outliers: Yes

Formula

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Example

Original Value (X)	Min (X_{min})	Max (X_{max})	Scaled Value
20	20	60	0.00
30	20	60	0.25
40	20	60	0.50
50	20	60	0.75
60	20	60	1.00

2 STANDARD SCALING (Z-SCORE SCALING)

Transforms features to have mean = 0 and standard deviation = 1.

- Range / Output: $-\infty$ to $+\infty$
- Best Used For: Regression, SVM, PCA, Logistic Regression
- Sensitive to Outliers: Yes

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Example

Original Value (X)	Mean (μ)	Std Dev (σ)	Scaled Value (Z)
20	40	14.14	-1.41
30	40	14.14	-0.71
40	40	14.14	0.00
50	40	14.14	0.71
60	40	14.14	1.41

3 ROBUST SCALING

Uses median and IQR, so it is robust to outliers.

- Range / Output: $-\infty$ to $+\infty$
- Best Used For: Data with Outliers
- Sensitive to Outliers: No

Formula

$$X_{scaled} = \frac{X - Median}{IQR}$$

where IQR = $Q_3 - Q_1$

Example

Original Value (X)	Median	IQR (Q_3-Q_1)	Scaled Value
10	30	20	-1.00
20	30	20	-0.50
30	30	20	0.00
40	30	20	0.50
50	30	20	1.00

4 MAX ABSOLUTE SCALING

Scales values by dividing with the maximum absolute value.

- Range / Output: -1 to 1
- Best Used For: Sparse Data, Text Data
- Sensitive to Outliers: Yes

Formula

$$X_{scaled} = \frac{X}{\max(|X|)}$$

Example

Original Value (X)	Max |X|	Scaled Value
-40	60	-0.67
-20	60	-0.33
0	60	0.00
20	60	0.33
60	60	1.00

5 UNIT VECTOR SCALING (NORMALIZATION)

Scales the vector to have a magnitude (length) of 1.

- Range / Output: Length = 1
- Best Used For: NLP, Cosine Similarity, Clustering
- Sensitive to Outliers: Less Sensitive

Formula

$$X_{scaled} = \frac{X}{||X||}$$

where $||X|| = \sqrt{\sum_{i=1}^{n} X_i^2}$

Original Vector [x_1, x_2, x_3]	Magnitude ($||X||$)	Scaled Vector
[3, 4, 0]	5	[0.60, 0.80, 0.00]
[1, 1, 1]	1.732	[0.58, 0.58, 0.58]

A. COMPARISON TABLE

Scaling Type	Output Range	Uses	Sensitive to Outliers	Typical Algorithms
Min-Max Scaling	0 to 1	KNN, Neural Networks, Image Processing	Yes	KNN, ANN, CNN
Standard Scaling (Z-Score)	$-\infty$ to $+\infty$	Regression, SVM, PCA	Yes	SVM, LR, PCA
Robust Scaling	$-\infty$ to $+\infty$	Data with Outliers	No	Tree Models, Robust Models
Max Absolute Scaling	-1 to 1	Sparse Data, Text Data	Yes	Linear Models, SVM
Unit Vector Scaling	Length = 1	NLP, Cosine Similarity, Clustering	Less Sensitive	K-Means, NLP

B. REAL-TIME APPLICATIONS

Banking	Used in credit scoring, loan approval prediction, fraud detection.
Healthcare	Helps in disease prediction, patient risk analysis, and treatment planning.
E-Commerce	Improves recommendation systems, customer segmentation, and sales forecasting.
Education	Used in student performance prediction, placement analysis, and learning analytics.
Weather Forecasting	Improves prediction models for temperature, rainfall, humidity, and storms.

C. KEY POINTS

- Feature Scaling is applied after data cleaning and before model training.
- It ensures that features contribute equally.
- Improves speed, accuracy, and stability of machine learning models.
- Always choose scaling method based on data distribution and algorithm.

D. MEMORY TIP

- N → Normalize (Min-Max)
- S → Standardize (Z-Score)
- R → Robust (Outliers)
- M → Max Absolute
- U → Unit Vector

KEY TAKEAWAY: Choose the right scaling based on data distribution, outliers, and algorithm type.
Right scaling leads to better learning, faster convergence, and higher accuracy!

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PART 5

7. DATA INTEGRATION & DATA AGGREGATION

1. DATA INTEGRATION

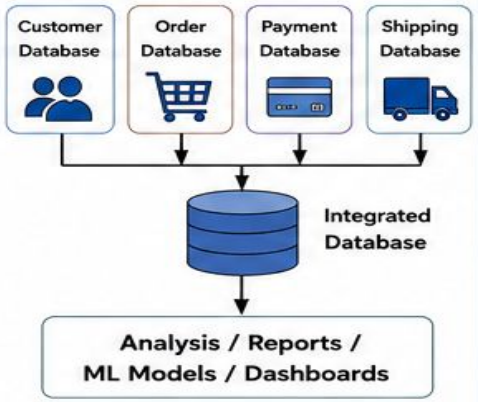
Meaning

Data Integration is the process of combining data from multiple sources into a single, unified and consistent dataset.

Definition

It is the process of merging data from different databases, files, applications, or systems to create one consistent view for analysis and decision-making.

Architecture



2. REAL-TIME EXAMPLES

A. Banking – 360° Customer View

- Core Banking System (Accounts)
- Loan Management System (Loans)
- Card System (Credit/Debit Cards)
- CRM System (Customer Interactions)

Result: Complete customer profile for credit scoring and personalized offers.

B. Healthcare – Patient Information System

- Patient Registration System
- Laboratory Information System
- Radiology System
- Billing & Insurance System

Result: Full patient history for better diagnosis and treatment.

C. Education – Student Information System

- Admission Records
- Examination System
- Attendance System
- Library System

Result: Complete student profile for performance analysis and placement.

D. E-Commerce – Unified Platform

- Customer Data
- Orders Data
- Payments Data
- Shipping Data

Result: Better customer experience, recommendations and sales analysis.

3. CHALLENGES IN DATA INTEGRATION

Duplicate Records	Same data appears in multiple sources with different IDs.
Different Formats (CSV, XLS, DB)	Data comes in CSV, Excel, JSON, Databases, APIs, etc.
Inconsistent Naming (ABC abc)	Same column names are different (e.g., Cust_ID, CustomerID, CID).
Data Conflicts	Different values for the same field (e.g., different addresses).
Missing Values	Some sources may have incomplete information.
Integration Complexity	Large number of sources increases time, cost, and complexity.

4. DATA AGGREGATION

Meaning

Data Aggregation is the process of summarizing data by combining multiple records into meaningful information.

Example

Consider sales data of a store:

Date	Product	Sales (₹)
01-06-2025	Mobile	15000
01-06-2025	Mobile	12000
01-06-2025	Laptop	30000
01-06-2025	Laptop	20000

Aggregated Result (Total Sales by Product)

Product	Total Sales (₹)
Mobile	27000
Laptop	50000

Common Aggregate Functions

Function	Description	Example	Result
SUM()	Total value	SUM(Sales)	77000
COUNT()	Number of records	COUNT(Product)	4
AVG()	Average value	AVG(Sales)	19250
MAX()	Highest value	MAX(Sales)	30000
MIN()	Lowest value	MIN(Sales)	12000

Aggregate functions help in summarizing large datasets and extracting valuable insights quickly.

5. BENEFITS

Better Analysis	Provides a complete and unified view for in-depth analysis.
Complete Information	All related data is combined, reducing information gaps.
Improved Decision-Making	Helps managers take accurate and timely decisions.
Time Saving	No need to check multiple systems separately.
Better Reporting	Helps generate consistent and meaningful reports.

6. APPLICATIONS

Banking	Customer profiling, fraud detection, risk management.
Healthcare	Patient care, medical research, insurance claim processing.
Education	Student performance tracking, learning analytics.
E-Commerce	Personalized recommendations, inventory and sales analysis.
Logistics	Route optimization, delivery tracking, supply chain analysis.
Social Media	User behavior analysis, targeted advertising.

KEY TAKEAWAY

Data Integration combines data from different sources into a single view, and Data Aggregation summarizes the data to produce meaningful information for better analysis, reporting, and decision-making.

MEMORY TIP

INTEGRATE to JOIN data,

AGGREGATE to SUMMARIZE data.

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PART 6

COMPLETE SUMMARY & END-TO-END FLOW

1. END-TO-END FLOW DIAGRAM

1. **Raw Data** — Unprocessed data from various sources
2. **Data Collection** — Collect data from databases, APIs, files, sensors, etc.
3. **Data Cleaning** — Remove duplicates, fix errors, standardize data
4. **Missing Value Handling** — Fill missing values using mean, median, mode or algorithms
5. **Data Transformation** — Normalize, standardize, encode, aggregate, create new values
6. **Feature Engineering** — Create, select, extract and encode meaningful features
7. **Feature Scaling** — Scale features to a common range (Min-Max, Z-Score, etc.)
6. **Data Integration** — Combine data from multiple sources into unified dataset
9. **Data Aggregation** — Summarize data to generate meaningful insights
10. **Machine Learning Model** — Train ML model with prepared data
11. **Prediction & Decision Making** — Generate predictions and support business decisions

2. COMPARISON TABLE

Aspect	Feature Engineering	Feature Scaling	Data Integration	Data Aggregation
Meaning	Create or modify features from existing data	Adjust numerical features to same scale	Combine data from multiple sources	Summarize data to generate insights
Purpose	Improve model accuracy by creating useful features	Ensure all features contribute equally	Get complete view by merging data	Reduce data size and provide summaries
Examples	Age from DOB, Income Ratio, Season, Encoding	Min-Max Scaling, Z-Score Scaling	Customer + Order + Payment + Shipping	Total Sales by Product, Average Marks by Subject
Effect on Model	Better accuracy, more informative data	Faster convergence, stable models	Complete information, better analysis	Better reporting and faster decision-making
When Used	Before model training after cleaning	After transformation before model training	When data comes from multiple sources	For reporting, dashboards, summary insights

3. MEMORY TRICKS

FEED the Model Better
- F – Feature Engineering (Create Smart Features)
- E – Enhance Accuracy
- E – Extract Important Info
- D – Deliver Better Predictions

SCALE Makes All Equal
- S – Same Range
- C – Contributes Equally
- A – Algorithm Works Well
- L – Learning Faster
- E – Enhanced Accuracy

INTEGRATE to UNITE Data
- I – Import from Many Sources
- N – Normalize and Clean
- T – Transform if Needed
- E – Ensure Consistency
- G – Get Unified View
- R – Reliable Analysis
- A – Accurate Decisions
- T – Trusted Information
- E – Effective Results

AGGREGATE to SUMMARIZE
- A – Add / Combine
- G – Group Data
- G – Generate Insights
- R – Reduce Size
- E – Easy to Understand
- G – Great for Reports
- A – Accurate Summary
- T – Timely Decisions
- E – Effective Outcomes

4. INTERVIEW QUESTIONS

1. What is Feature Engineering? Why is it important?
2. What are the different types of Feature Engineering?
3. What is Feature Scaling? Why is it required?
4. What are the types of Feature Scaling?
5. What is the difference between Normalization and Standardization?
6. What is Data Integration? Why is it needed?
7. What are the challenges in Data Integration?
8. What is Data Aggregation? Give real-time examples.
9. How do these techniques improve ML model performance?

5. VIVA QUESTIONS

1. Define Feature Engineering with an example.
2. How do you handle categorical data?
3. When do you apply Feature Scaling?
4. Which scaling method is best for outliers?
5. What is the difference between Min-Max Scaling and Z-Score Scaling?
6. What is the need of Data Integration in real life?
7. What is the difference between Data Integration and Data Aggregation?
8. Explain with an example each.

6. PYTHON (GOOGLE COLAB) EXAMPLES

A. FEATURE ENGINEERING

```
import pandas as pd
# Sample Data
df = pd.DataFrame({
    'DOB': ['2000-01-10','1998-05-20'],
    'Salary': [50000, 80000],
    'Loan': [200000, 100000]
})

from datetime import datetime
df['Age'] = df['DOB'].apply(
    lambda x: 2025 - datetime.strptime(x, '%Y-%m-%d').year)
# Loan to Income Ratio
df['Loan_to_Income_Ratio'] = df['Loan'] / df['Salary']
df
```

B. FEATURE SCALING (MIN-MAX)

```
from sklearn.preprocessing import MinMaxScaler
import pandas as pd

data = pd.DataFrame({
    'Age': [20, 30, 40, 50],
    'Salary': [30000, 50000, 70000, 90000]
})

scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(data)
scaled_df = pd.DataFrame(scaled_data,
                         columns=data.columns)
scaled_df
```

C. DATA INTEGRATION

```
# Combine Data from Different Sources
import pandas as pd

cust = pd.DataFrame({'Cust_ID':[1,2],'Name':['Ram','Sita']})
orders = pd.DataFrame({'Cust_ID':[1,2],'Order_ID':[101,102]})
pay = pd.DataFrame('Order_ID':[101,102],
                   'Amount':[2000,1500]})

# Merge all
integrated = cust.merge(orders, on='Cust_ID')
                 .merge(pay, on='Order_ID')
integrated
```

D. DATA AGGREGATION

```
# Sales Data Aggregation
import pandas as pd

data = pd.DataFrame({
    'Date': ['01-06-2025']*4,
    'Product': ['Mobile','Mobile','Laptop','Laptop'],
    'Sales': [15000, 12000, 30000,20000]
})

agg = data.groupby('Product')['Sales'].sum().reset_index()
agg.rename(columns={'Sales': 'Total_Sales'}, inplace=True)
agg
```

7. REAL-WORLD APPLICATIONS (SUMMARY)

- **Banking** — Loan Prediction, Fraud Detection, Customer 360° View
- **Healthcare** — Disease Prediction, Patient Analysis, Hospital Management
- **Education** — Student Performance, Placement Prediction, Learning Analytics
- **E-Commerce** — Recommendations, Sales Forecasting, Customer Insights
- **Logistics** — Route Optimization, Delivery Tracking, Supply Chain Analysis
- **Social Media** — User Behavior Analysis, Sentiment Analysis, Targeted Ads

8. KEY TAKEAWAY

Feature Engineering creates smart features from raw data.
Feature Scaling ensures all features contribute equally.
Data Integration combines data from multiple sources.
Data Aggregation summarizes data to generate insights.
Together, they improve data quality, model accuracy, and decision-making in real-world applications.

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